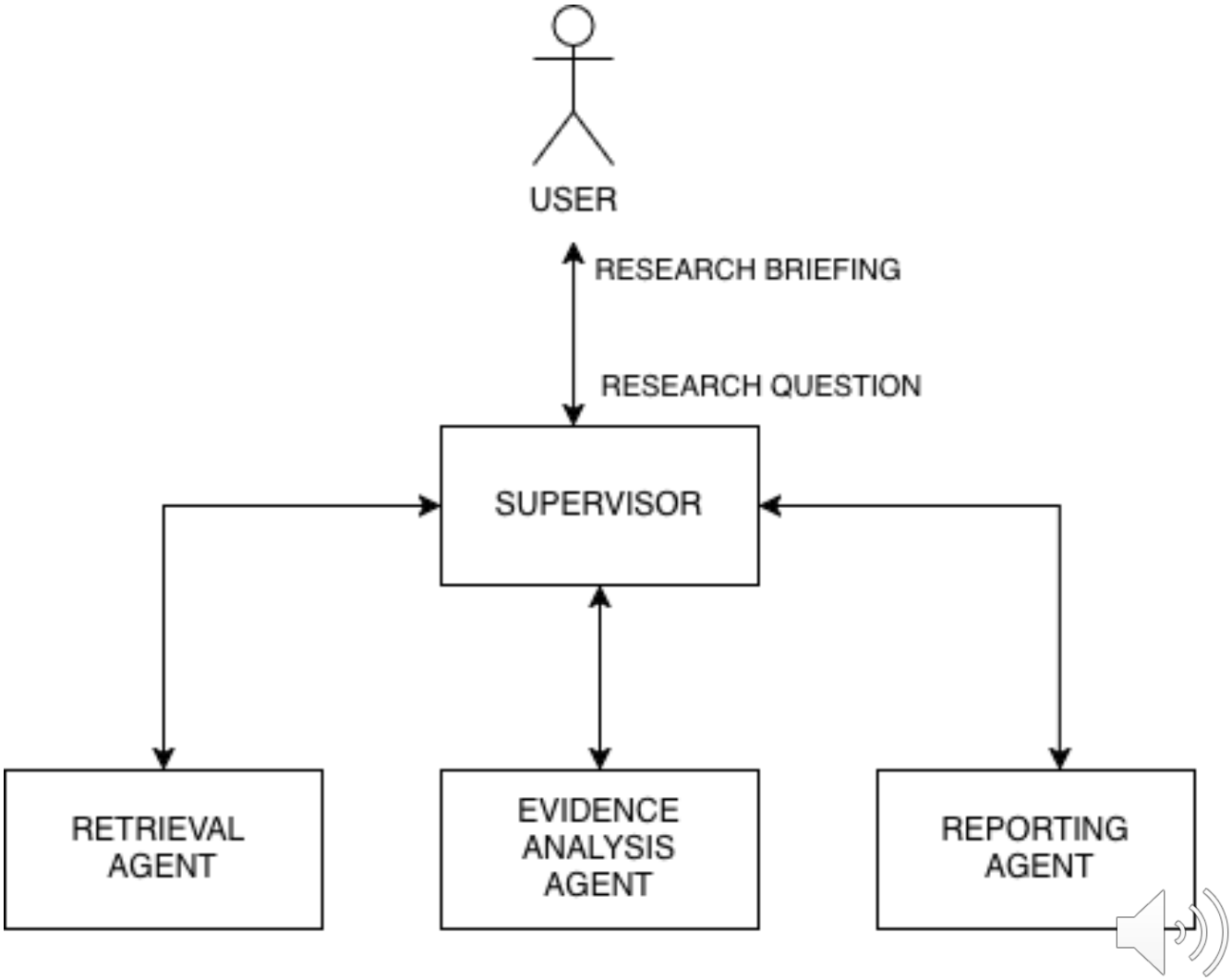
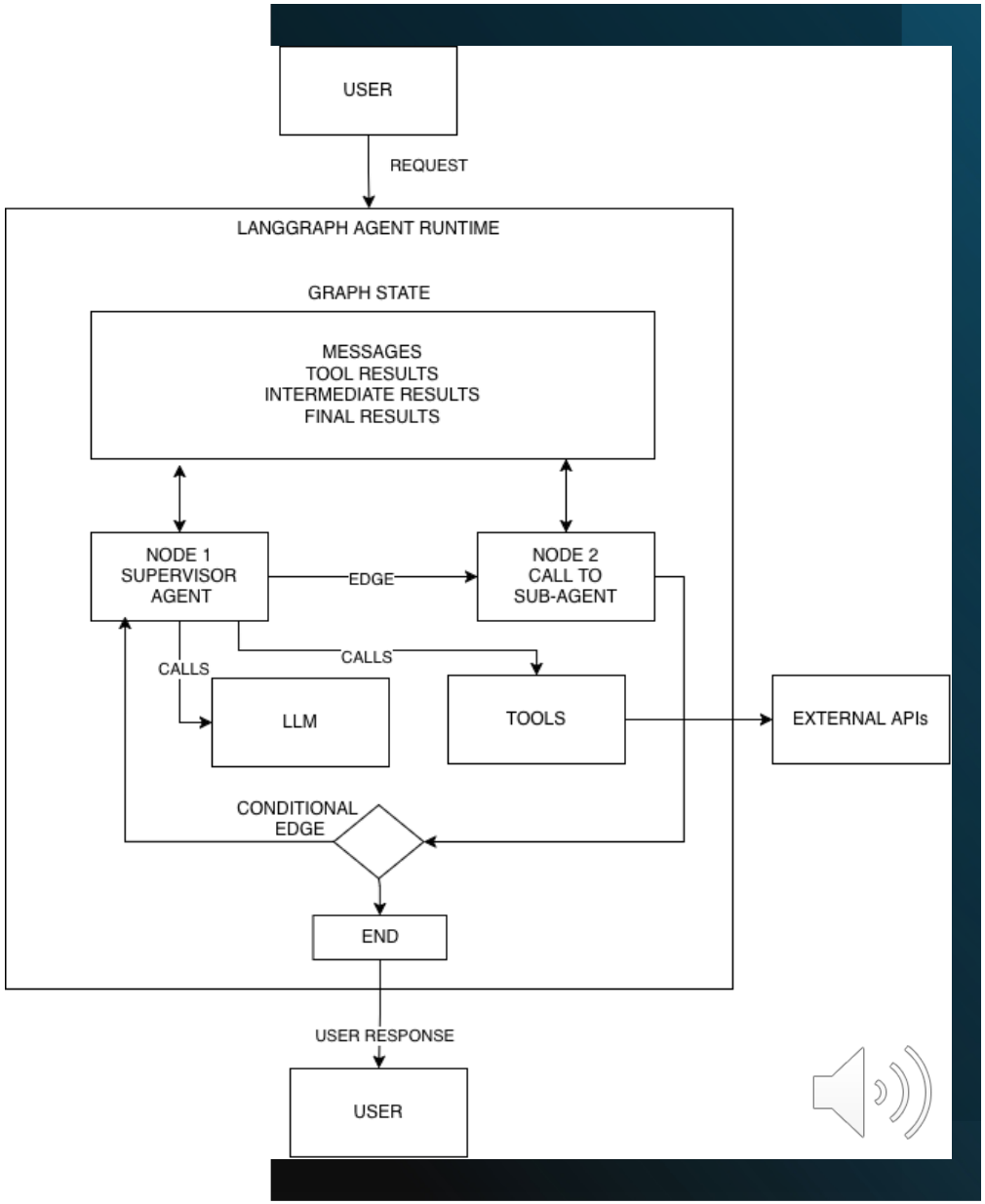
Intelligent Multi-Agent Academic Investigator

Logical Architecture. Supervisor pattern

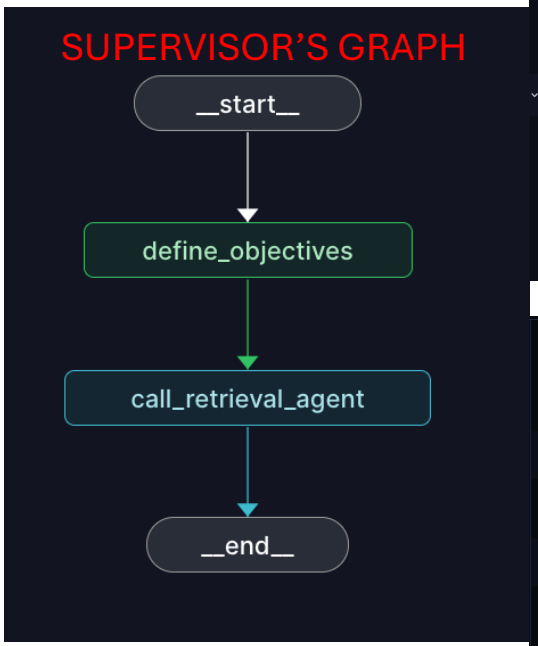


Runtime and Technological Architecture

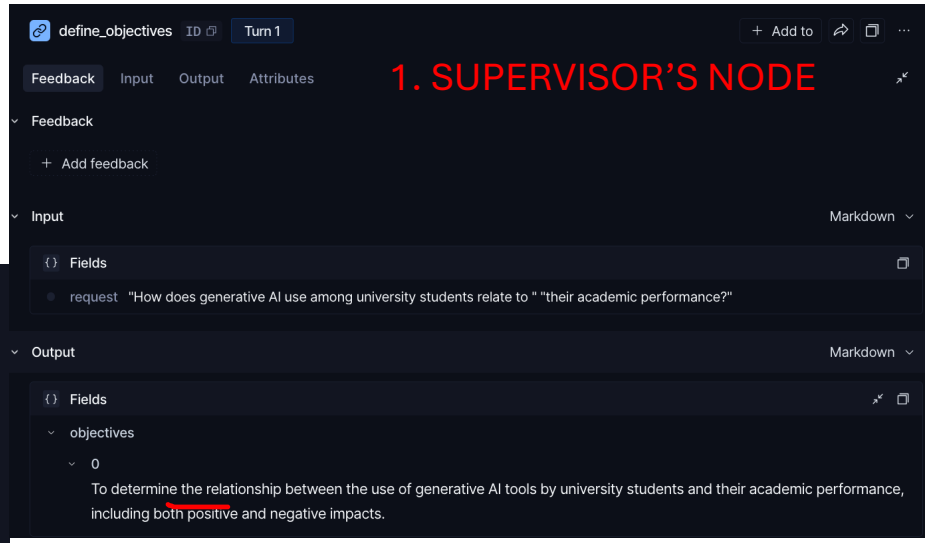
- LangGraph
  - State
  - Nodes
  - Edges
- LLM
  - Qwen3:14b over Ollama
  - Reliably deployed on a 16GB workstation (Song et al., 2025)
  - Strong performance in information extraction (OCUL, 2026)
- Tools
  - Semantic Scholar's Academic Graph API
    - Large academic database
    - Free use for academic institutions

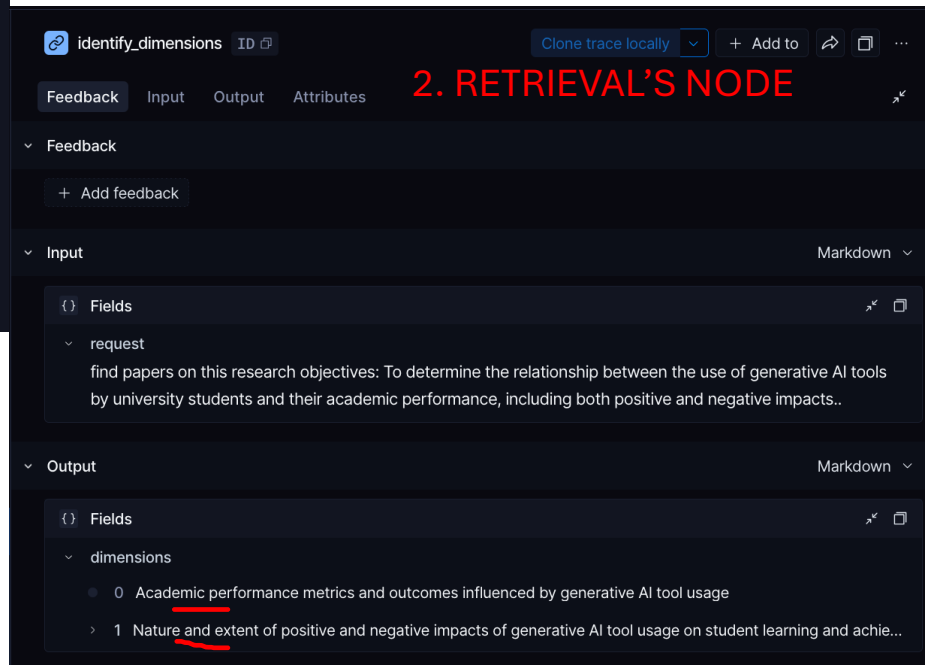


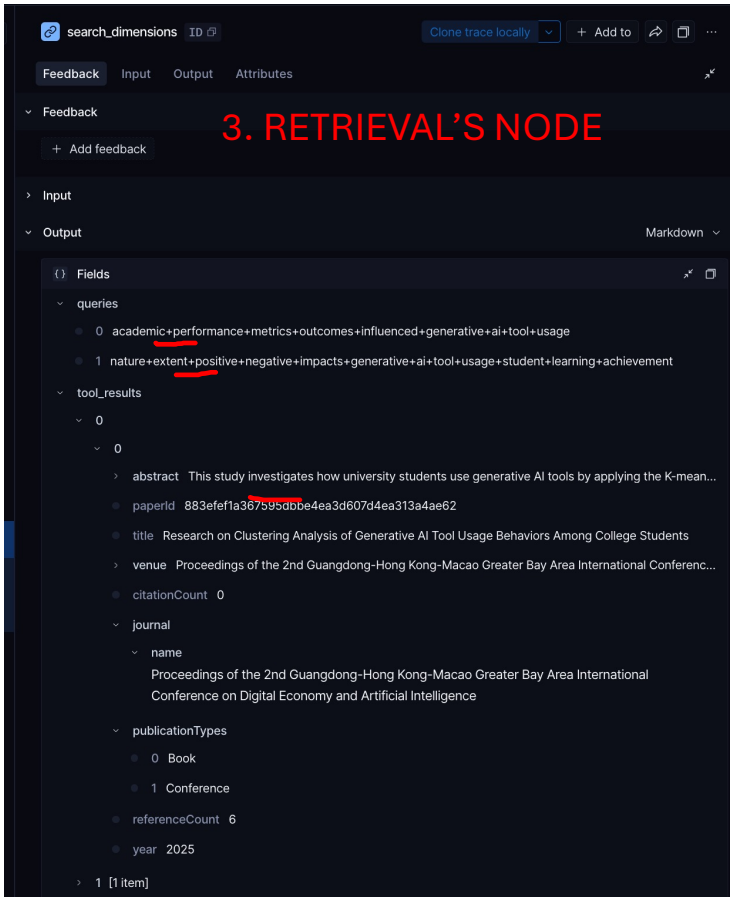
V1.0.0



Research objectives translates a broader research question into explicit targets (Creswell & Creswell, 2017).

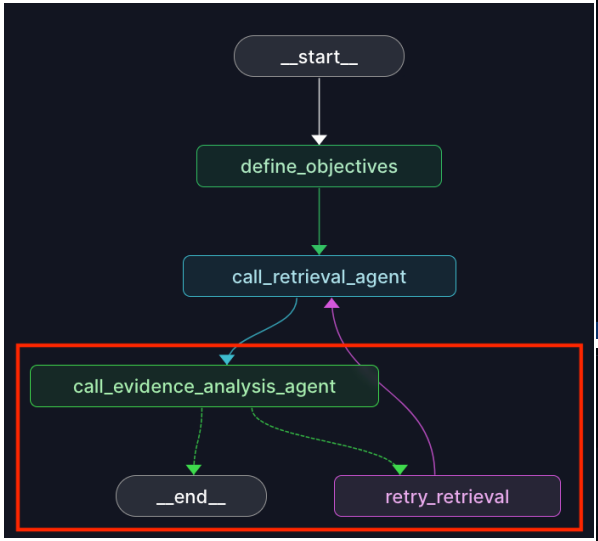


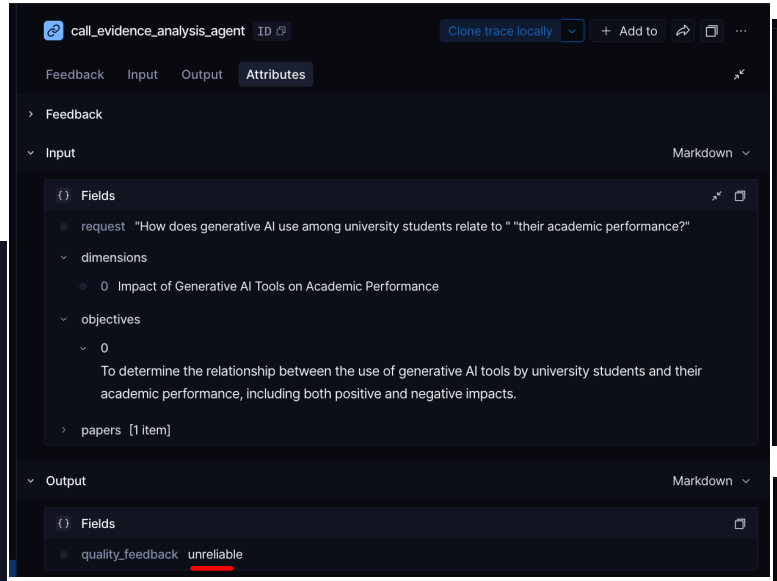


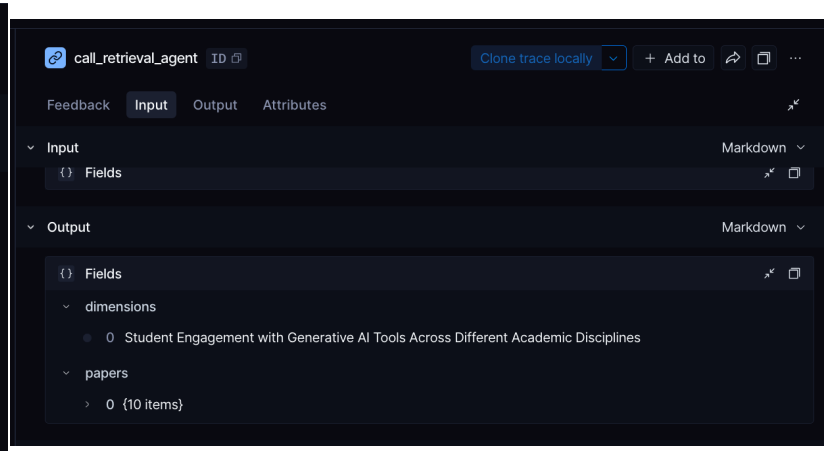


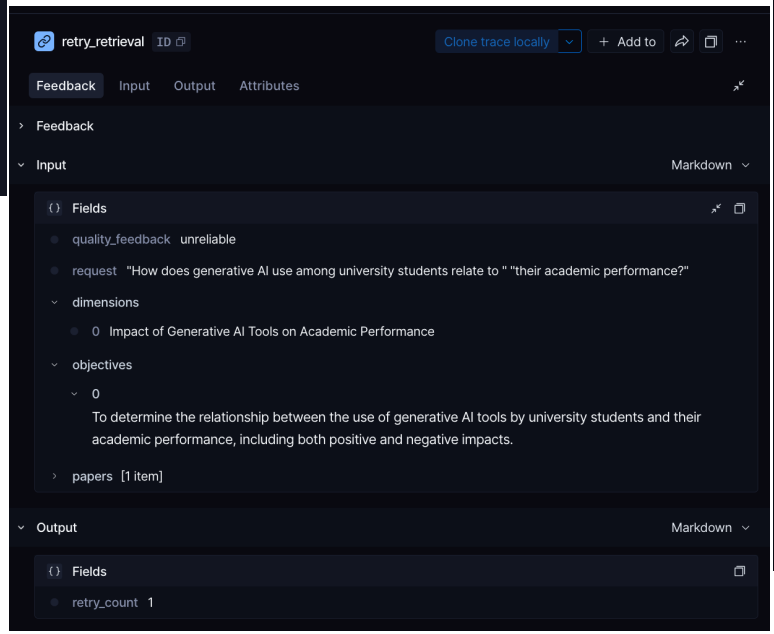


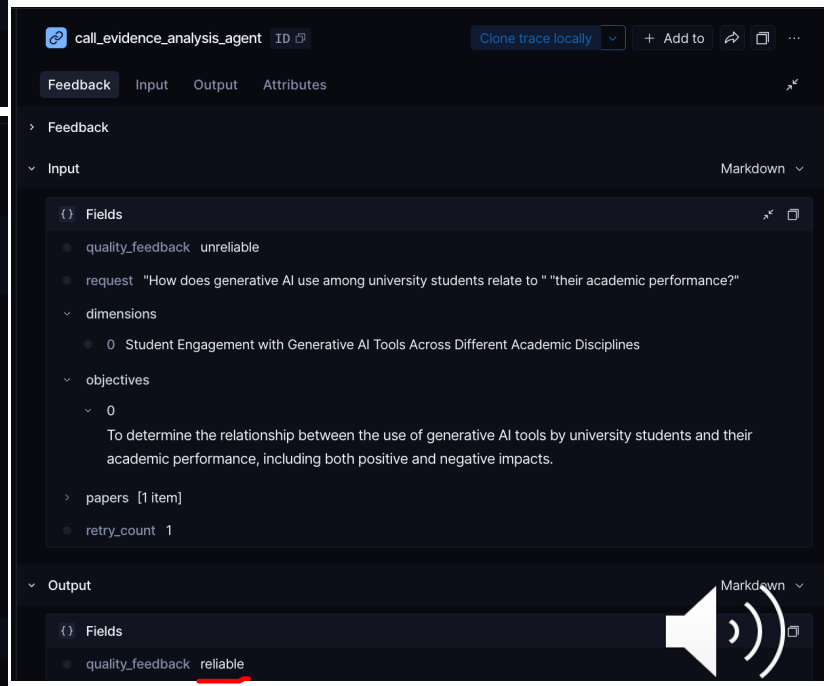
V2.0.0


__start__
define_objectives
call_retrieval_agent
call_evidence_analysis_agent
__end__
retry_retrieval

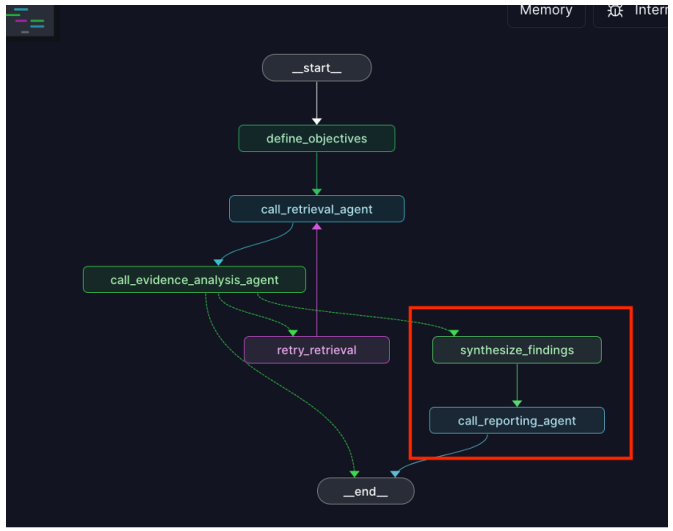

call_evidence_analysis_agent
ID
Clone trace locally
Add to
Feedback
Input
Output
Attributes
Feedback
Input
Markdown
Fields
request "How does generative AI use among university students relate to " "their academic performance?"
dimensions
0 Impact of Generative AI Tools on Academic Performance
objectives
0
To determine the relationship between the use of generative AI tools by university students and their academic performance, including both positive and negative impacts.
papers [1 item]
Output
Markdown
Fields
quality_feedback unreliable

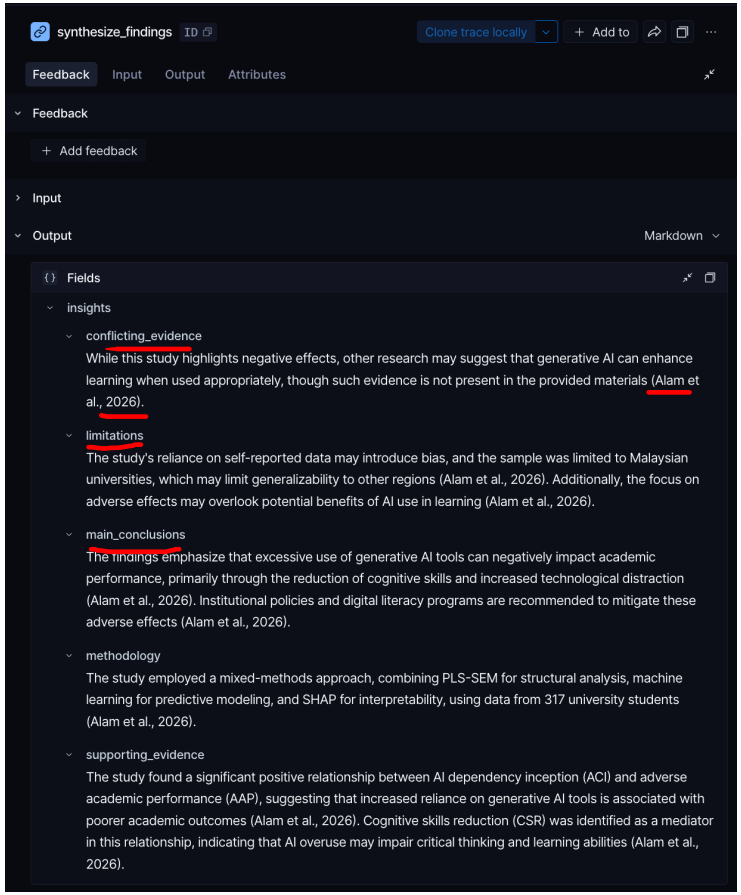

call_retrieval_agent
ID
Clone trace locally
Add to
Feedback
Input
Output
Attributes
Input
Markdown
Fields
Output
Markdown
Fields
dimensions
0 Student Engagement with Generative AI Tools Across Different Academic Disciplines
papers
0 {10 items}

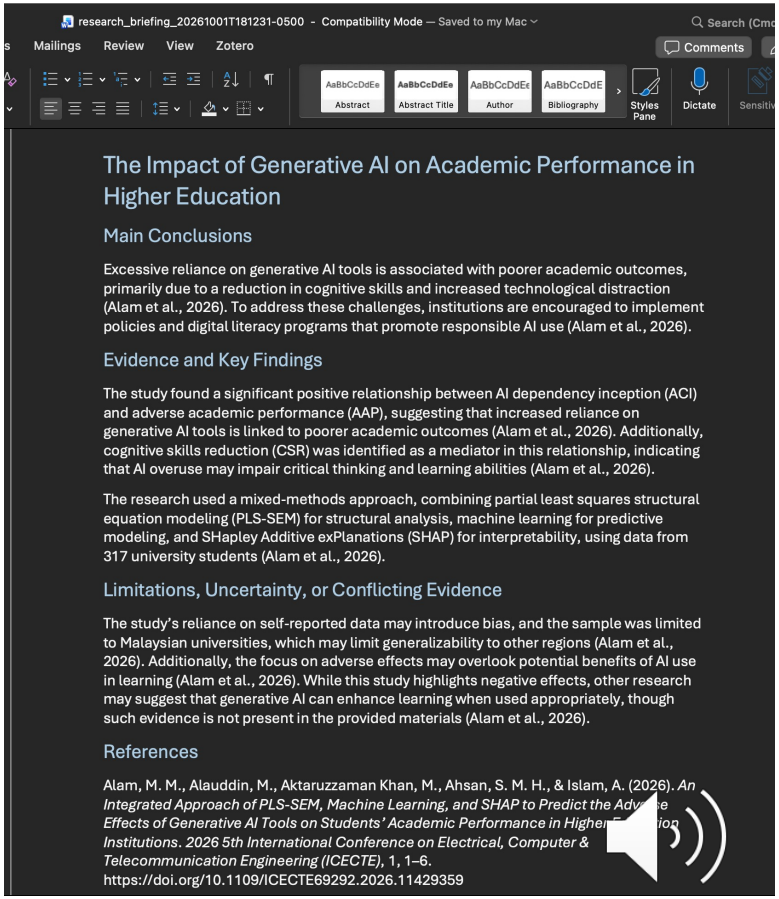

retry_retrieval
ID
Clone trace locally
Add to
Feedback
Input
Output
Attributes
Feedback
Input
Markdown
Fields
quality_feedback unreliable
request "How does generative AI use among university students relate to " "their academic performance?"
dimensions
0 Impact of Generative AI Tools on Academic Performance
objectives
0
To determine the relationship between the use of generative AI tools by university students and their academic performance, including both positive and negative impacts.
papers [1 item]
Output
Markdown
Fields
retry_count 1


call_evidence_analysis_agent
ID
Clone trace locally
Add to
Feedback
Input
Output
Attributes
Feedback
Input
Markdown
Fields
quality_feedback unreliable
request "How does generative AI use among university students relate to " "their academic performance?"
dimensions
0 Student Engagement with Generative AI Tools Across Different Academic Disciplines
objectives
0
To determine the relationship between the use of generative AI tools by university students and their academic performance, including both positive and negative impacts.
papers [1 item]
retry_count 1
Output
Markdown
Fields
quality_feedback reliable

V3.0.0


Memory
__start__
define_objectives
call_retrieval_agent
call_evidence_analysis_agent
retry_retrieval
synthesize_findings
call_reporting_agent
__end__


synthesize_findings
ID
Clone trace locally
Add to
Feedback
Input
Output
Attributes
Feedback
Add feedback
Input
Output
Markdown
Fields
insights
conflicting_evidence
While this study highlights negative effects, other research may suggest that generative AI can enhance learning when used appropriately, though such evidence is not present in the provided materials (Alam et al., 2026).
limitations
The study's reliance on self-reported data may introduce bias, and the sample was limited to Malaysian universities, which may limit generalizability to other regions (Alam et al., 2026). Additionally, the focus on adverse effects may overlook potential benefits of AI use in learning (Alam et al., 2026).
main_conclusions
The findings emphasize that excessive use of generative AI tools can negatively impact academic performance, primarily through the reduction of cognitive skills and increased technological distraction (Alam et al., 2026). Institutional policies and digital literacy programs are recommended to mitigate these adverse effects (Alam et al., 2026).
methodology
The study employed a mixed-methods approach, combining PLS-SEM for structural analysis, machine learning for predictive modeling, and SHAP for interpretability, using data from 317 university students (Alam et al., 2026).
supporting_evidence
The study found a significant positive relationship between AI dependency inception (ACI) and adverse academic performance (AAP), suggesting that increased reliance on generative AI tools is associated with poorer academic outcomes (Alam et al., 2026). Cognitive skills reduction (CSR) was identified as a mediator in this relationship, indicating that AI overuse may impair critical thinking and learning abilities (Alam et al., 2026).


research_briefing_20261001T181231-0500 - Compatibility Mode — Saved to my Mac
Mailings
Review
View
Zotero
Comments
Abstract
Abstract Title
Author
Bibliography
Styles Pane
Dictate
The Impact of Generative AI on Academic Performance in Higher Education
Main Conclusions
Excessive reliance on generative AI tools is associated with poorer academic outcomes, primarily due to a reduction in cognitive skills and increased technological distraction (Alam et al., 2026). To address these challenges, institutions are encouraged to implement policies and digital literacy programs that promote responsible AI use (Alam et al., 2026).
Evidence and Key Findings
The study found a significant positive relationship between AI dependency inception (ACI) and adverse academic performance (AAP), suggesting that increased reliance on generative AI tools is linked to poorer academic outcomes (Alam et al., 2026). Additionally, cognitive skills reduction (CSR) was identified as a mediator in this relationship, indicating that AI overuse may impair critical thinking and learning abilities (Alam et al., 2026).
The research used a mixed-methods approach, combining partial least squares structural equation modeling (PLS-SEM) for structural analysis, machine learning for predictive modeling, and SHapley Additive exPlanations (SHAP) for interpretability, using data from 317 university students (Alam et al., 2026).
Limitations, Uncertainty, or Conflicting Evidence
The study's reliance on self-reported data may introduce bias, and the sample was limited to Malaysian universities, which may limit generalizability to other regions (Alam et al., 2026). Additionally, the focus on adverse effects may overlook potential benefits of AI use in learning (Alam et al., 2026). While this study highlights negative effects, other research may suggest that generative AI can enhance learning when used appropriately, though such evidence is not present in the provided materials (Alam et al., 2026).
References
Alam, M. M., Alauddin, M., Aktaruzzaman Khan, M., Ahsan, S. M. H., & Islam, A. (2026). An Integrated Approach of PLS-SEM, Machine Learning, and SHAP to Predict the Adverse Effects of Generative AI Tools on Students' Academic Performance in Higher Education Institutions. 2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), 1, 1–6. https://doi.org/10.1109/ICECTE69292.2026.11429359

Implementation decision 1

- Fully deterministic search tool (Semantic Scholar API)
- Advantages
  - Reliable
  - Efficient
- Limitations
  - Hard-coded retrieval strategy for any research domain
- Solution
  - LLM dynamically retrieves relevant papers through a Semantic Scholar API skill.
  - Domain-aware Search strategy.

NODE

```python
def search_dimensions(state: RetrievalState) -> dict:
    """Step 2: build one query per dimension, invoke search_papers once per query, and extract the usable papers."""
    queries = [_dimension_to_query(dimension) for dimension in state["dimensions"]]
    print("Searching identified dimensions with tool")
    tool_results = [search_papers.invoke({"query": query}) for query in queries]      You, last week • move analyse fi
    print(f"Completed search for {len(queries)} queries")
    papers = _extract_papers(tool_results)
    if not papers:
        raise RuntimeError("No paper titles or abstracts were available. Try another research question.")
    return {"queries": queries, "tool_results": tool_results, "papers": papers}
```

TOOL

```python
@tool
def search_papers(
    query: str
) -> list[Paper]:
    """Search for academic papers using the given query."""
    print(f"Searching for papers with query: {query}")
    query_params = {
        "query": query,
        "limit": 1,
        "fields": "paperId,title,abstract,year,referenceCount,citationCount,publicationTypes,journal,venue,authors,
        externalIds"
    }
```



Implementation decision 2

NODE

Simple Evidence criticism

- Only bibliometric indicators such as recency and citation metrics are used

Advantages

- Fast to implement

Limitations

- It overlook methodological weaknesses, bias risk on individual evidence
- Evidence set might lack of diversity.
  - Narrow set of perspectives
- Evidence set might lack of balance.
  - E.g., Disproportionately supportive/opposing,

Solution

- Multi-dimensional evidence criticism
  - Add criticism on the evidence set level
  - Assess diversity and balance

```
You, 2 weeks ago | author (You)
class EvidenceAnalysisState(TypedDict):
    papers: list[Paper]
    quality_feedback: str

""" implementation decision: make evidence quality assessment fully deterministic to
 economize token usage and reduce response time. """
def assess_evidence_quality_node(state: EvidenceAnalysisState, config: RunnableConfig) -> dict:
    """Call the assess_evidence_quality tool directly for each paper, without going through an LLM."""
    papers = state.get("papers") or []
    if not papers:
        raise RuntimeError("No papers were provided to assess evidence quality.")

    print(f"Assessing evidence quality for {len(papers)} papers")
    tool_results = [assess_evidence_quality.invoke({"paper": paper}, config=config) for paper in papers]
    if all(r == RELIABLE_DESCRIPTION for r in tool_results):
        quality_feedback = RELIABLE_DESCRIPTION
    elif all(r == UNRELIABLE_DESCRIPTION for r in tool_results):
        quality_feedback = UNRELIABLE_DESCRIPTION
    else:      You, 2 weeks ago • integrated evidence analysis with supervisor …
        quality_feedback = QUESTIONABLE_DESCRIPTION
    print(f"Evidence quality assessment for {len(papers)} papers completed with feedback: {quality_feedback}")
    return {"quality_feedback": quality_feedback}
```

Testing

- Pytest framework

	What it checks	Dependencies	How I used it	Example
Unit tests	Single agent's internal logic	Removed dependencies by Pytest's Monkeypatch	On each agent's change	test_retrieval_agent.py
Integration tests	Single agent's integration with external tools and LLMs	LangGraph, LLM, APIs	On LLM prompt changes, tool changes	test_reporting_agent_integration.py
Functional tests	End-to-end validation of functional requirements	LangGraph, LLM, APIs, other agents	On major changes involving multiple agents	test_supervisor_functional.py



Test report and coverage

```
============================================================================ tests coverage ============================================================================
_____________________________________________________ coverage: platform darwin, python 3.14.3-final-0 _____________________________________________________

Name                                      Stmts   Miss  Cover   Missing
-----------------------------------------------------------------------
src/agents/evidence_analysis_agent.py        70      2    97%   26, 82
src/agents/reporting_agent.py                59      4    93%   72, 109, 121-122
src/agents/retrieval_agent.py                77      5    94%   26-28, 42, 68
src/agents/supervisor.py                     98      3    97%   147, 156, 219
src/main.py                                  14     14     0%   2-27
src/tools/search_api.py                      89     18    80%   31, 109-110, 112-122, 127-132, 137
src/utils/framework.py                       15      2    87%   27, 35
src/utils/logging_config.py                  17      0   100%
-----------------------------------------------------------------------
TOTAL                                       439     48    89%
========================================================================== slowest durations ===========================================================================
96.32s call     tests/test_supervisor_functional.py::TestSupervisorGraph::test_full_graph_succeeds_with_reliable_evidence_on_first_try
81.71s call     tests/test_supervisor_functional.py::TestSupervisorGraph::test_full_graph_succeeds_with_unreliable_evidence_on_first_try_then_retry
64.31s call     tests/test_reporting_agent_integration.py::TestReportingAgentIntegration::test_briefing_generation
3.68s call     tests/test_retrieval_agent_integration.py::TestManageSearch::test_search_covers_multiple_dimensions_using_mock_endpoint_server_with_success_response
2.27s call     tests/test_retrieval_agent_integration.py::TestManageSearch::test_search_covers_multiple_dimensions_using_mock_endpoint_server_with_error_response
0.50s teardown tests/test_retrieval_agent_integration.py::TestManageSearch::test_search_covers_multiple_dimensions_using_mock_endpoint_server_with_success_response
0.49s teardown tests/test_retrieval_agent_integration.py::TestManageSearch::test_search_covers_multiple_dimensions_using_mock_endpoint_server_with_error_response
0.27s teardown tests/test_supervisor_functional.py::TestSupervisorGraph::test_full_graph_succeeds_with_unreliable_evidence_on_first_try_then_retry
0.03s call     tests/test_manage_evidence_analysis.py::TestManageEvidenceAnalysis::test_assesses_quality_of_gathered_papers_as_reliable
0.02s setup    tests/test_retrieval_agent_integration.py::TestManageSearch::test_search_covers_multiple_dimensions_using_mock_endpoint_server_with_success_response
0.01s setup    tests/test_supervisor_functional.py::TestSupervisorGraph::test_full_graph_succeeds_with_reliable_evidence_on_first_try
0.01s call     tests/test_retrieval_agent.py::TestIdentifyDimensions::test_returns_dimensions_from_structured_output

(51 durations < 0.005s hidden.  Use -vv to show these durations.)
========================================================================== 21 passed in 249.96s (0:04:09) ==========================================================================
(.venv) luistorres@Luiss-MacBook-Air mas-academic-investigator %
```



DEMONSTRATION

The complete code and documentation are available at https://github.com/luitoz/multi-agent-academic-investigator

REFERENCES

- Creswell, J.W. and Creswell, J.D. (2017) *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. SAGE Publications.
- Ontario Council of University Libraries (OCUL) (2026) 'AI for Academic Libraries: Open-Weight AI Models for Local and Private Use', *Choice 360*, 8 June. Available at: https://www.choice360.org/libtech-insight/ai-for-academic-libraries-open-weight-ai-models-for-local-and-private-use/ (Accessed: 3 October 2026).
- Song, Q. *et al.* (2025) 'A Systematic Evaluation of On-Device LLMs: Quantization, Performance, and Resources'. arXiv. Available at: https://doi.org/10.48550/ARXIV.2505.15030.